

Quadratic Condensation and Relational Local Time in a Many-Wave Complete Relational System — From Relational Closure without Spacetime as Fundamental Variables to the Kepler Equation and the Inverse-Square Central Force

Noriaki Kihara (WF System Co., Ltd.) ORCID 0009-0004-6753-4020

September 16, 2026 Version DOI 10.5281/zenodo.22788738

Abstract

This study reports an unexpected conservation structure obtained from numerical experiments that extend the finite complete relational closed system of our preceding work to a nonlinear system containing many relational waves. The initial states were generated as linear sums of two integer harmonic components; the dynamical kernel was never given the harmonic numbers or the parity classification, and all states were updated simultaneously by a real antisymmetric generator constructed only from the relative phases between relational waves sharing a vertex. Running the preregistered 99 runs for 4096 steps, the anticipated relaxation-type evolution — a long stay on the floor, an onset, and saturation — did not occur: 5 runs are exact fixed points, and the remaining 94 depart from the initial plane almost immediately.

On the other hand, across all runs and all times,

$$\sum_m z_m^2 = C^2$$

was confirmed to be conserved as one and the same complex constant C^2 . This follows rigorously from the fact that the update matrix is real orthogonal. In this paper we read this identical conservation structure as **quadratic condensation**: many internal waves are represented by a single complex quadratic quantity. For each condensate n , defining

$$C_n^2 = \sum_m z_{nm}^2,$$

we have

$$\sum_n C_n^2 = \sum_{n,m} z_{nm}^2,$$

so that at the higher level one may again impose

$$\sum_n C_n^2 = 0.$$

Quadratic closure therefore closes hierarchically in the same form even after the internal degrees of freedom are condensed.

Furthermore, the simultaneous conservation of $H = Z^\dagger Z$ and $C^2 = Z^T Z$ is equivalent to fixing the Gram matrix of the real and imaginary parts of the state, which determines the eigenvalues $\lambda_\pm = (H \pm |C^2|)/2$ and the shape index $\chi = |C^2|/H \in [0, 1]$. Under nonvanishing components and connectivity, $\chi = 1 \iff K = 0$ holds; in the measurements, exactly the 5 exact fixed points among the 99 runs have $\chi = 1$ (all others have

$\chi \leq 0.486$). For a partition into condensates, an antisymmetric quadratic flux $J_{nm} = 2Z_n^T K_{nm} Z_m = -J_{mn}$ is defined, with $\dot{C}_n^2 = \sum_{m \neq n} J_{nm}$ — local conservation, pairwise exchange, and global closure are connected within a single structure. In addition, for the two-mode initial family, χ is determined by a closed form in (L, m_a, m_b) , so the local Gram shape is decided before any time evolution is executed.

For the minimal two-condensate system

$$C_1^2 + C_2^2 = 0,$$

one obtains $C_2 = \pm i C_1$, a 90-degree phase relation. The transformations preserving this closure can be defined as real orthogonal transformations, and in the continuous representation as conservative motion generated by a real antisymmetric generator. No external physical time needs to be assumed as a fundamental variable. When two local condensates A, B exist, B is a clock for A and A is a clock for B through their relative phase; no absolute clock is required. A single-valued relational phase $\tilde{U}_{AB}$ is determined from the quadratic representatives alone, and the two lifts on its double cover give the two time directions $U_{AB} = \pm i$ at minimal closure. A complete derivation of time itself is a task for the next stage. What this paper shows is that a long internal state sequence closes within one invariant quadratic sector, and that local clocks can be defined as inter-system relations.

Furthermore, squaring the fixed Gram 2-frame yields an ellipse with one focus at the origin, whose eccentricity coincides exactly with $\chi = |C^2|/H$. In the three-component embedding with the complementary component $i\sqrt{|C^2|}$, any nondegenerate elliptic eccentricity $0 \leq e < 1$ is realized while exactly preserving quadratic zero closure, with $C^2 = 0$ corresponding to the circular orbit. Postulating the clock-readout axiom $dt = r d\tau = |u|^2 d\tau$, the Kepler equation for the eccentric anomaly appears, and the same transformation yields the constancy of the areal velocity and the inverse-square central force $\ddot{\mathbf{r}} = -\mu \mathbf{r}/r^3$. Thus in this system the central structure of Kepler dynamics is reconstructed without placing physical time or spatial coordinates as fundamental variables at the outset.

As for inflation-type relaxation, it is highly likely that the present 99 runs sampled only the two extremes: exact fixed points and low-symmetry immediate departure. Comparison with the relaxation curves observed in past high-symmetry 90-degree configurations suggests, as a candidate, an “intermediate window”: high symmetry, slightly off the fixed point, with the centroid near the origin. The verification procedure is given in the conclusion.

Keywords: complete relational system, quadratic condensation, quadratic closure, fixed Gram geometry, shape index chi, hierarchical closure, antisymmetric quadratic flux, relational local time, double cover, squaring map, Kepler equation, inverse-square central force

1. Introduction

In the preceding study [K1], without placing external time as a distinguished physical variable from the outset, states at different phase instants were treated as ordinary state vertices, and the minimal exactly closed system whose complete network of relations closes was analyzed. In particular, for

$$U^K = I$$

with $N = 1, K = 4$, using the base wave and one integer harmonic

$$S_k = (U^k, U^{mk}),$$

the orders, complete-relation distances, and quadratic closures of even versus odd harmonics were compared exactly.

Its direct background, the preceding study [K2], placed

$$M = \frac{N(N-1)}{2}$$

complex relational waves on the complete graph over N vertices; iterating the nonlinear interaction from near a highly symmetric 90-degree phase series, the waves were observed numerically to pass from the initial cross-shaped band, through a rapid growth of the transverse component, to a quasi-stable ring of nearly equal amplitudes with widely dispersed phases.

The original motivation of the present work was simple: to extend the two-wave structure of paper 1 [K1] to complete relational systems with many more relational waves, and to look for general structure among even/odd harmonics, intrinsic periods, finite cycles, and long-time evolution.

The outcome differed from the expectation. Even after running the 99 runs to 4096 steps, the intended relaxation-type temporal structure was not obtained. Instead, a simpler and stronger identity spanning all runs appeared:

$$\boxed{\sum_m z_m^2 = C^2}.$$

The purpose of this paper is to organize this result in the following stages.

1. **Numerical fact:** In the nonlinear evolution of the 99 runs, $\sum_m z_m^2$ was conserved as one complex constant.
2. **Mathematical consequence:** The many internal waves can be exactly contracted to a single quadratic representative C^2 , and this contraction is closed under hierarchization.
3. **Candidate physical interpretation:** From the relative phase relations of multiple condensates, local clocks can possibly be defined without assuming an absolute clock.
4. **Geometric consequence:** Kepler ellipses are derived from the squaring map of the fixed Gram geometry (eccentricity $e = \chi$).
5. **Dynamical consequence:** Under $dt = |u|^2 d\tau$ obtained from the clock-readout axiom, the Kepler equation and the inverse-square central force are reconstructed.

Items 1, 2, 4, and 5 above are treated as mathematical/numerical results; item 3 is distinguished as interpretation and as a future verification task.

2. Numerical experimental system

2.1 Initial states

For a finite cyclic length L , set

$$U = \exp\left(\frac{2\pi i}{L}\right).$$

Let the undirected edges of the complete graph be $e = (j, k)$, $j < k$, and use

$$\Delta = k - j.$$

For two mode numbers m_a, m_b , the initial state is

$$z_{jk}(0) = c_a U^{m_a \Delta} + c_b U^{m_b \Delta},$$

where

$$c_a = e^{-i\pi/(2L)}, \quad c_b = e^{+i\pi/(2L)}.$$

The initial state is thus a **simple superposition of just two finite phase waves**. The mode numbers are used only for generating the initial condition and are never passed to the subsequent dynamics.

2.2 Complete relational dynamics

Two edges e, f are adjacent when they share a vertex; let A_{ef} be the adjacency matrix of the line graph. Writing each state as

$$z_e = |z_e|e^{i\theta_e},$$

construct the real antisymmetric generator

$$K_{ef}(Z) = A_{ef} \sin(\theta_f - \theta_e).$$

The one-step update is

$$Z' = \exp(\Delta\tau K(Z))Z.$$

Since

$$K^T = -K,$$

the matrix

$$R(Z) = \exp(\Delta\tau K(Z))$$

is a real orthogonal matrix at every step.

K is state dependent, so the dynamics as a whole is nonlinear, but **each simultaneous update itself is a real orthogonal transformation.**

2.3 The 99-run survey

The preregistered survey consists of 99 runs in total, all storing the 4097 states

$$t = 0, \dots, 4096.$$

Table 1: Composition of the 99 runs

Series	L	Mode pairs	den	Runs
Main survey	8,10,12,16,20,24,32,40	(1,3),(1,5),(3,5),(2,4),(2,6),(4,6),(1,2),(2,3),(3,6)	L	72
gcd series	12,16,20,24,32,40	(3,9),(4,8),(5,10)	L	18
Step-size controls	12,24,32	(1,3),(2,4),(2,3)	40	9
Total				99

All 99 runs passed QA, and the raw data were never modified by post-processing.

3. Catalog of initial states: the geometry generated by the two-wave generator

The catalog (Figure 1) is the completed figure that overlays, for the same 99 initial states, the point configuration, the two-component phasor composition paths with multiplicities, the centroid, and the quadratic representative. The subsections below correspond to the elements of Figure 1.

Figure 1a. Catalog of initial states (completed figure): main survey, 72 runs

Figure 1b. Catalog of initial states (completed figure): gcd series 18 + den=40 controls 9

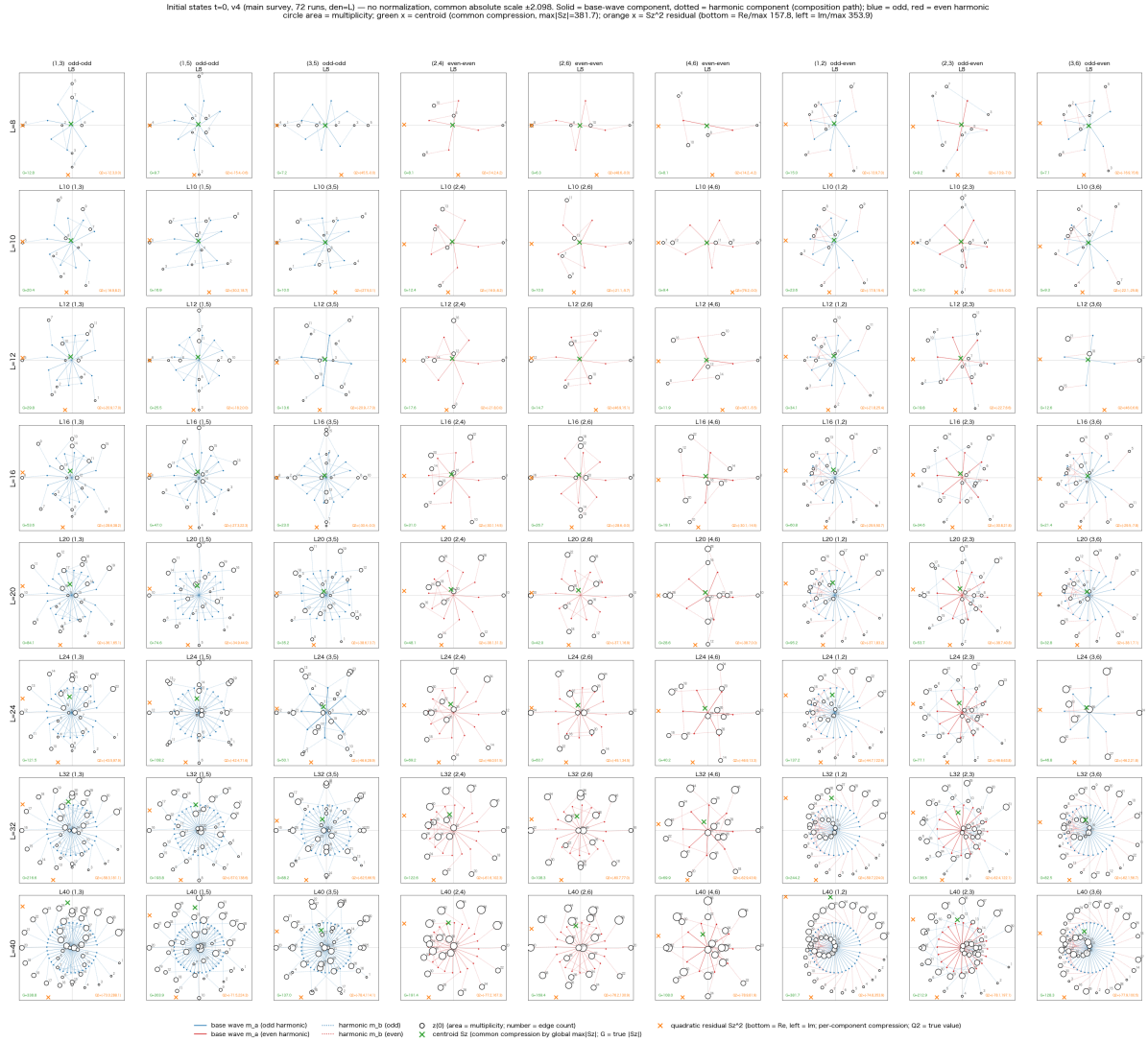


Figure 1: Figure 1a initial states catalog, main 72

Initial states t=0, v4 (gcd series 18 runs, top 6 rows + den=40 controls 9 runs, bottom 3 rows)
 No normalization, common absolute scale ± 2.098 . Solid = base wave, dotted = harmonic; blue = odd, red = even
 circle area = multiplicity; green x = centroid (common compression, $\max(|S_i|)=391.7$)
 orange x = Sz^2 residual (bottom = $\text{Re}/\max 157.8$, left = $\text{Im}/\max 353.9$)

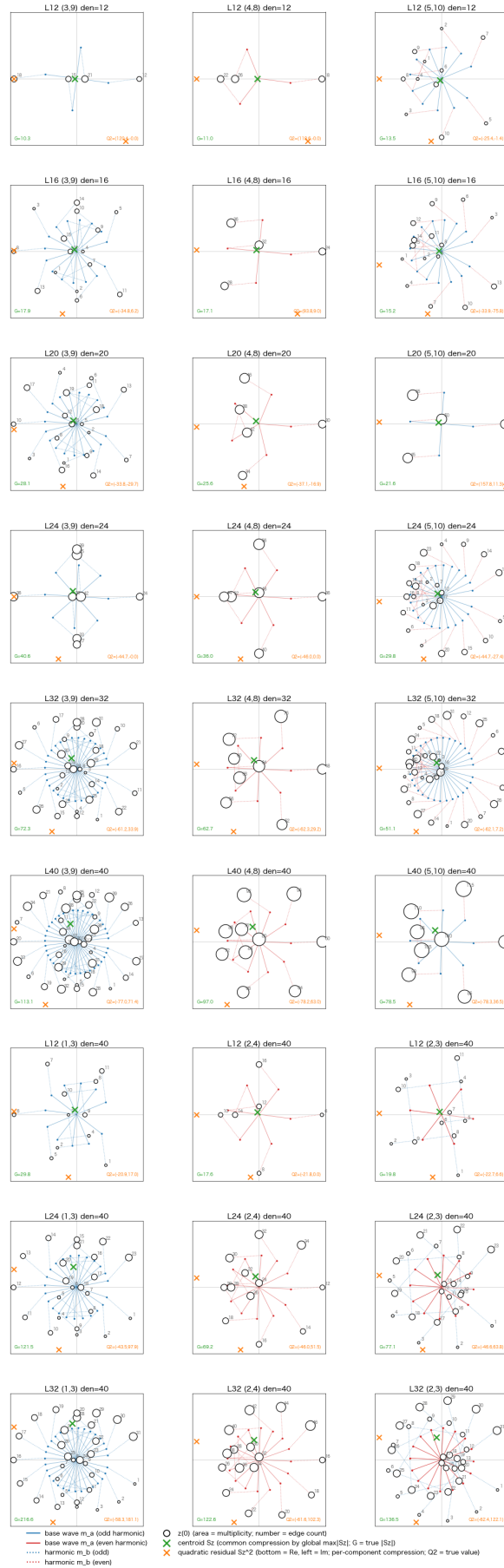


Figure 2: Figure 1b initial states catalog, gcd+control

3.1 Point configuration

The initial state depends only on $\Delta = k - j$, so many edges degenerate to the same point. All edges are nonzero, and as L grows, characteristic configurations — stars, crosses, sparse layouts, double rings — appear according to the mode pair.

3.2 Two-component composition paths and multiplicities

The composition paths in Figure 1 (solid = base-wave component, dotted = harmonic component) make explicit that each initial complex number $z_e(0)$ is generated as the sum of two mode components. However complicated the dynamics, the starting point is a simple two-component superposition.

3.3 Centroid

In all 99 runs the centroid

$$G = \sum_e z_e$$

was nonzero. The centroid is not conserved and reorganizes substantially during the evolution. This paper therefore does not regard the centroid as a conserved representative of a condensate but treats it as an **auxiliary quantity reading the degree of symmetric configuration**.

3.4 Quadratic quantity

At the time the frame marks of Figure 1 (real and imaginary parts of $\sum z^2$) were created, we called $\sum z^2$ the “quadratic-closure residual” and read it as a deviation from zero. The subsequent comparison of the full time series showed this reading to be insufficient.

That

$$\sum_e z_e^2 \neq 0$$

for all 99 runs is not by itself a “violation of closure”. Rather, for each run a

$$\boxed{\sum_e z_e^2 = C^2}$$

with a **run-specific complex constant** C^2 is selected, and the same value is conserved for all time.

Hereafter we therefore call this the **quadratic representative** rather than a residual.

4. Complete linear superposition of waveforms

Linearly superposing, over all M edges, the two generating waves carried by each edge shows directly which frequency components the initial state contains and with what gain.

Figure 2a. Complete linear superposition of all M edge waves: main 72

Figure 2b. Complete linear superposition of all M edge waves: gcd + controls

With the initial modes m_a, m_b , the total superposition can be written

$$W(x) = \Re \left[\left(\sum_e A_e \right) e^{im_a x} + \left(\sum_e B_e \right) e^{im_b x} \right].$$

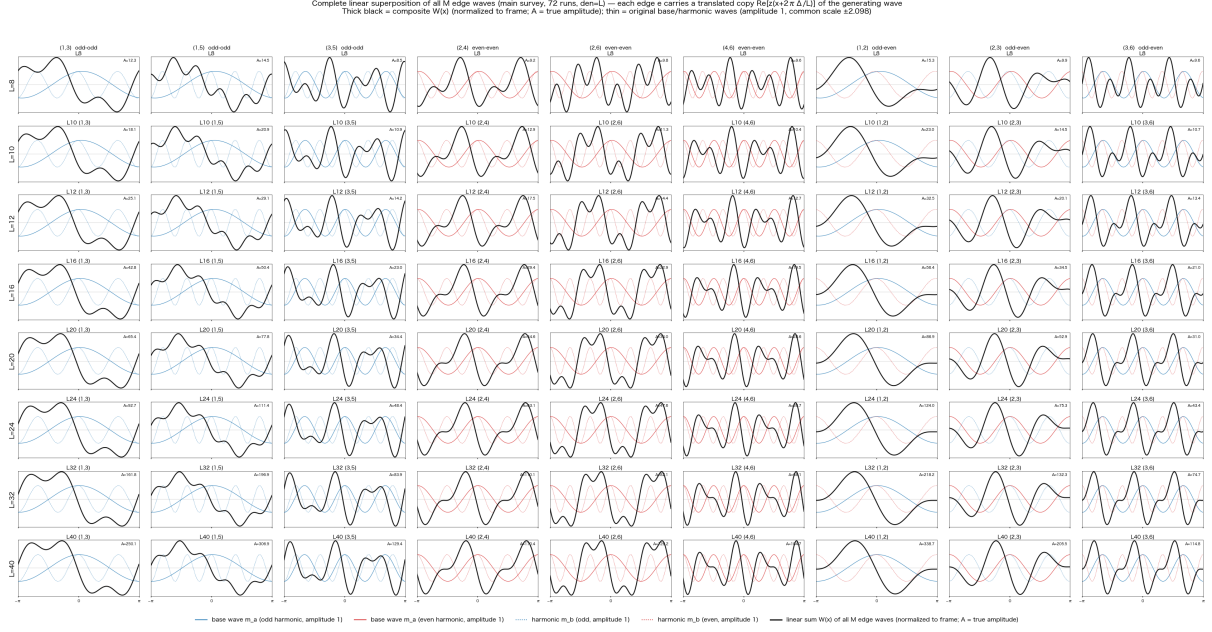


Figure 3: Figure 2a linear superposition, main 72

For small m/L the base-wave side often dominates, but in a finite cyclic system the folding of $\sin(\pi m/L)$ produces gain inversions unlike a simple continuous frequency ratio.

The initial configuration is two waves, but the subsequent nonlinear dynamics knows nothing of the mode labels. Hence the quadratic conservation obtained later is not the result of “conserving with the harmonic numbers held fixed”.

5. Final states: internal reorganization after 4096 steps

Figure 3a. Final states $t = 4096$: main 72

Figure 3b. Final states $t = 4096$: gcd + controls

In most runs with $L \geq 16$, the initially degenerate point configuration breaks up and reorganizes into ring-like distributions of nearly equal amplitude. On the other hand, the 5 runs satisfying

$$m_a + m_b \equiv 0 \pmod{L}$$

have initial states that are real vectors up to a common phase and remain exact fixed points with

$$K = 0,$$

preserving the initial configuration. These 5 runs are instances of the theorem of §6.3 ($\chi = 1 \iff K = 0$) and satisfy $\chi = 1$ exactly (Figure 5).

The internal state may therefore either stay still or reorganize substantially depending on the run. As the next section shows, in either case the quadratic representative C^2 is conserved in the same sense.

6. Central result: $\sum z_m^2 = C^2$ closes exactly for all time

6.1 Numerical observation

Figure 4a. Time series of $\sum z^2$ and centroid $\sum z$: main 72

Complete linear superposition of all M edge waves (gcd series 18 runs, top 6 rows + den=40 controls 9 runs, bottom 3 rows)
 Thick black = composite $W(x)$ (normalized to frame; A = true amplitude); thin = original waves (amplitude 1, ± 2.098)

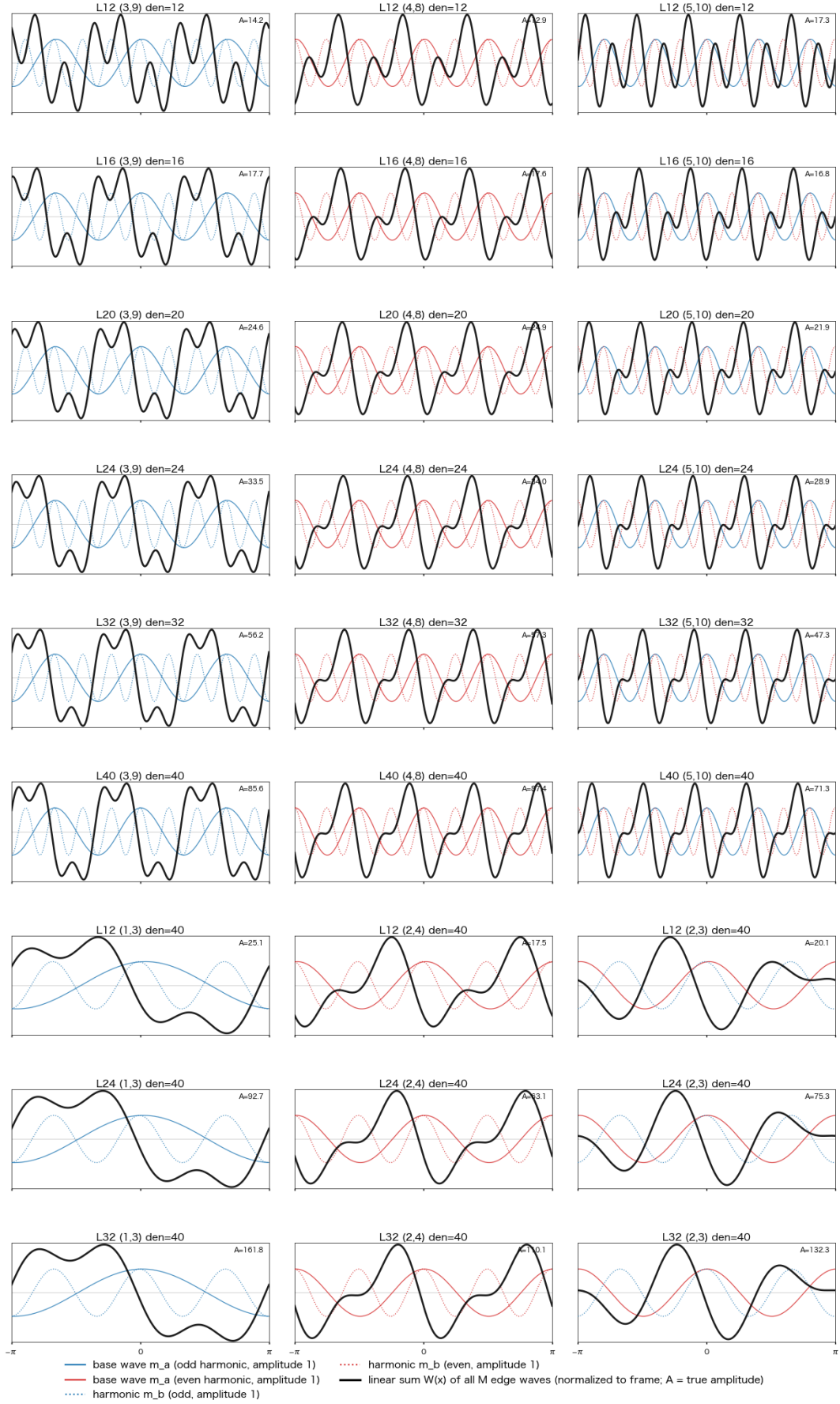


Figure 4: Figure 2b linear superposition, gcd+control

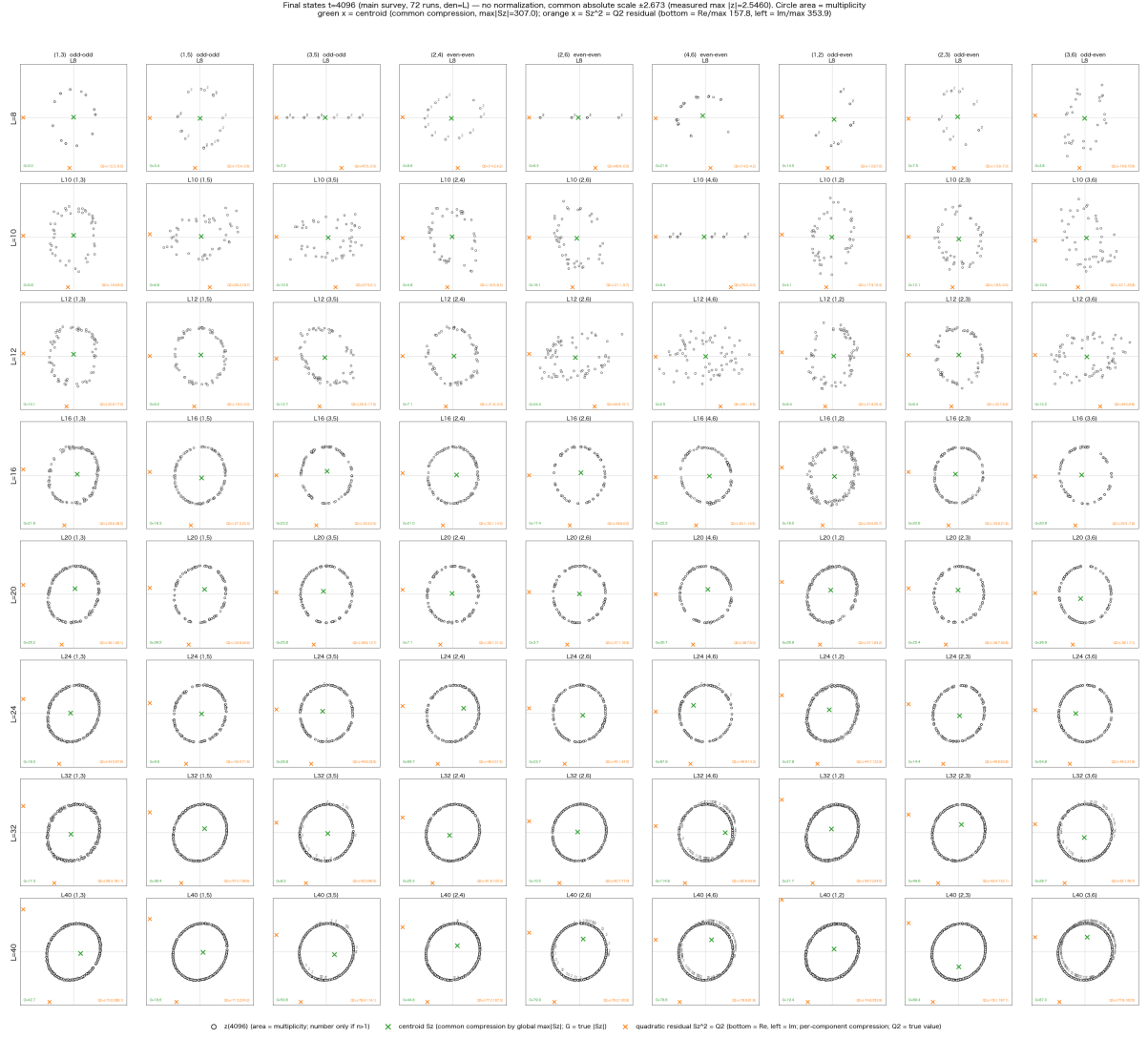


Figure 5: Figure 3a final states, main 72

Final states t=4096 (gcd series 18 runs, top 6 rows + den=40 controls 9 runs, bottom 3 rows)
 No normalization, common absolute scale ± 2.673 . Circle area = multiplicity
 green x = centroid (common compression, max|Sz|=307.0), orange x = Sz² = Q2 (bottom = Re, left = Im)

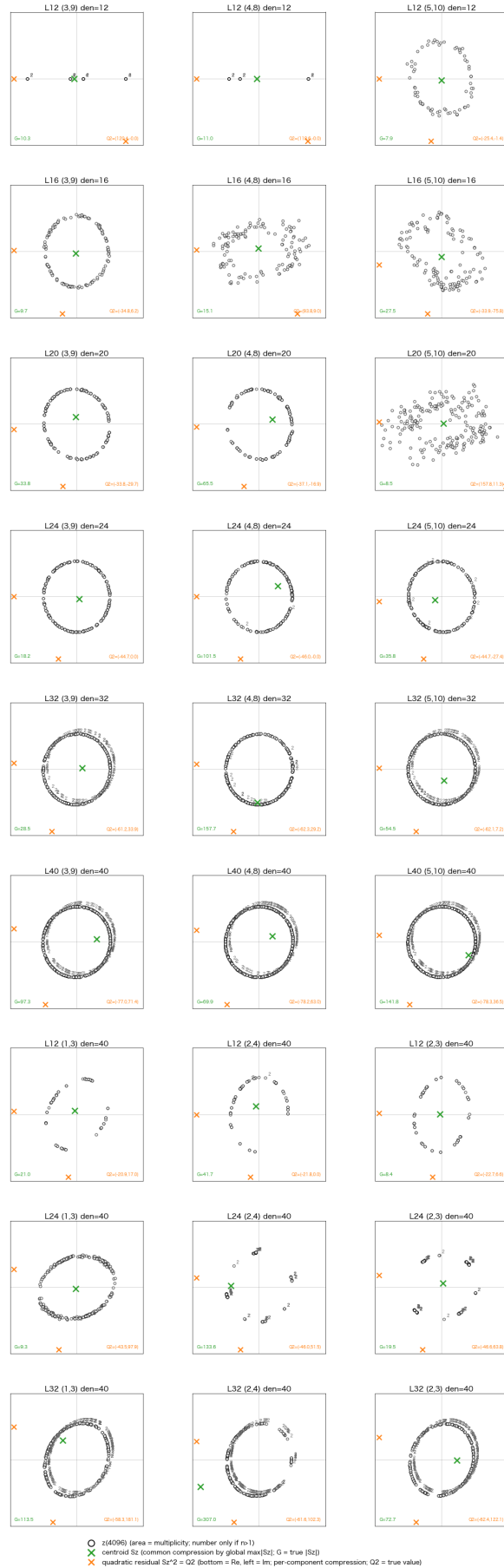


Figure 6: Figure 3b final states, gcd+control

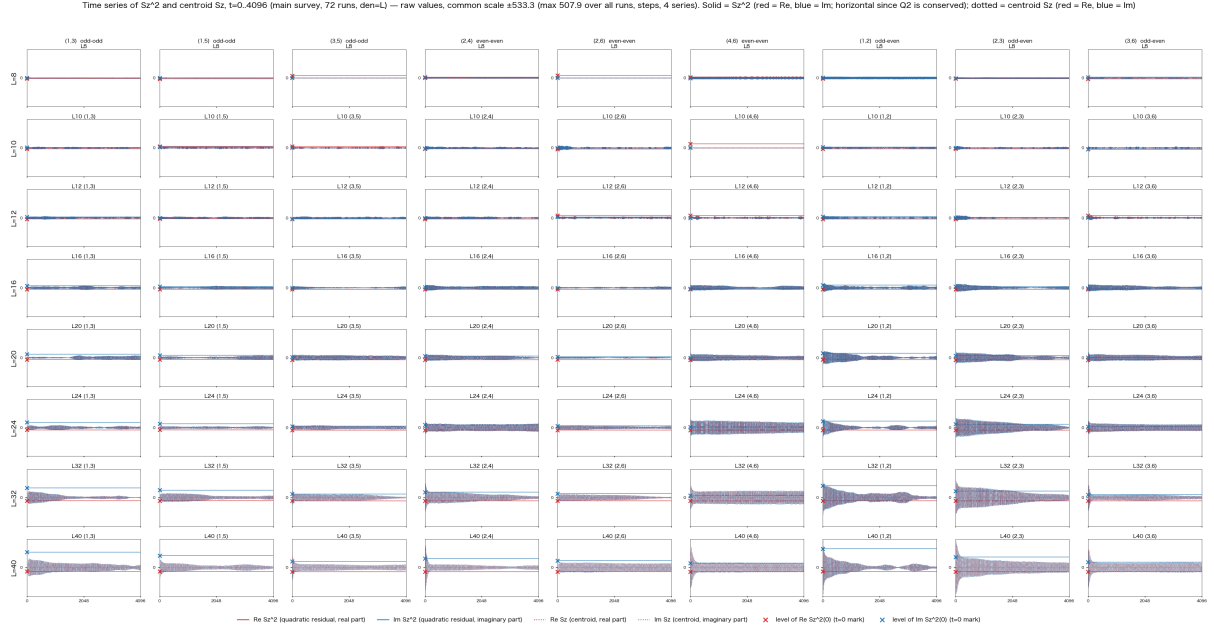


Figure 7: Figure 4a residual time series, main 72

Figure 4b. Time series of $\sum z^2$ and centroid $\sum z$: gcd + controls

In Figure 4, while the centroid $\sum z$ oscillates strongly and sometimes amplifies beyond its initial value,

$$Q_2(t) = \sum_m z_m(t)^2$$

is a horizontal line in every panel. In the numerical audit over all 99 runs with 4097 states each, the drift of Q_2 stayed within $\max_t |Q_2(t) - Q_2(0)| = 4.1 \times 10^{-10}$ (relative $\sim 10^{-12}$ against $H_0 \sim 10^3$).

What matters at this point is not the numerical approximation

$$Q_2(t) \approx Q_2(0)$$

but the following algebraic identity.

6.2 Derivation of the exact conservation

Write one step as

$$Z' = R(Z)Z, \quad R(Z) = e^{\Delta\tau K(Z)}.$$

Since $K^T = -K$,

$$R^T R = I.$$

Therefore

$$Z'^T Z' = Z^T R^T R Z = Z^T Z,$$

that is,

$$\sum_m z_m'^2 = \sum_m z_m^2.$$

Time series of Sz^2 and centroid Sz , $t=0.4096$ (gcd series 18 runs, top 6 rows + den=40 controls 9 runs, bottom 3 rows)
Raw values, common scale ± 533.3 . Solid = Sz^2 (red = Re, blue = Im; conserved); dotted = centroid Sz (red = Re, blue = Im)

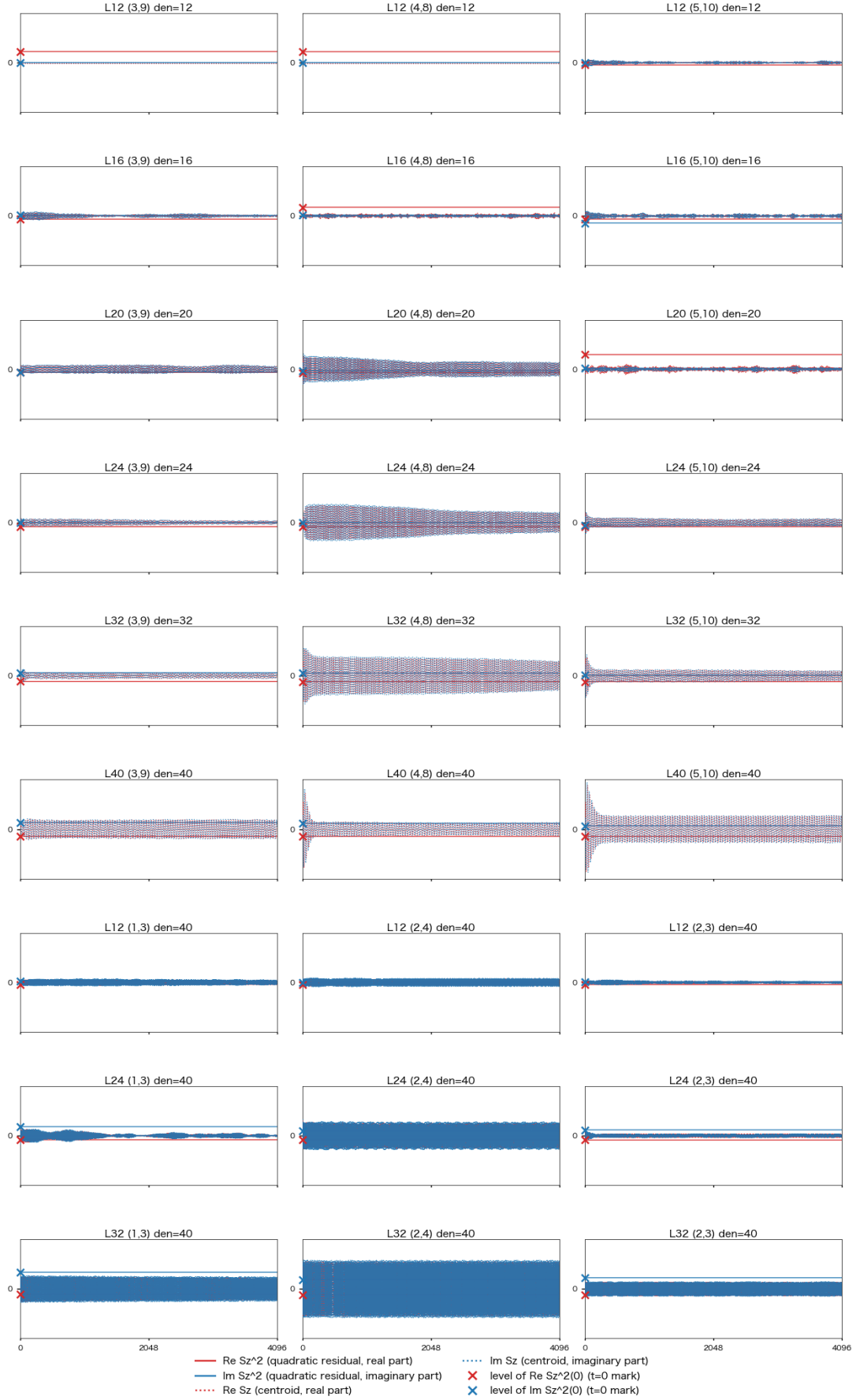


Figure 8: Figure 4b residual time series, gcd+control

Since R is a real matrix, $R^\dagger = R^T$, and simultaneously

$$\boxed{Z'^\dagger Z' = Z^\dagger R^T R Z = Z^\dagger Z = H}$$

also holds. Hence

$$\boxed{H = \text{const}, \quad C^2 = \text{const}}$$

is complete within this section alone.

Defining at the initial value

$$C^2 := \sum_m z_m(0)^2,$$

for any step t ,

$$\boxed{\sum_m z_m(t)^2 = C^2}$$

holds.

The dynamics is nonlinear through the state dependence of $K(Z)$, but the nonlinearity never moves the orbit to another quadratic sector. The entire orbit closes inside

$$\mathcal{M}_{C^2} = \{Z \in \mathbb{C}^M \mid Z^T Z = C^2\}.$$

6.3 Fixed Gram geometry and the shape index χ

Decomposing $Z = x + iy$ ($x, y \in \mathbb{R}^M$),

$$H = Z^\dagger Z = x^2 + y^2, \quad C^2 = Z^T Z = x^2 - y^2 + 2i x \cdot y,$$

and the simultaneous conservation of H and C^2 is equivalent to

$$x^2 = \frac{H + \Re C^2}{2}, \quad y^2 = \frac{H - \Re C^2}{2}, \quad x \cdot y = \frac{\Im C^2}{2},$$

that is, to fixing the Gram matrix of the real and imaginary parts,

$$G = \begin{pmatrix} x^2 & x \cdot y \\ x \cdot y & y^2 \end{pmatrix}.$$

The real orthogonal update $Z' = RZ$ acts as $x' = Rx$, $y' = Ry$, so this is a derivation, not an additional assumption. The eigenvalues of G are

$$\boxed{\lambda_{\pm} = \frac{H \pm |C^2|}{2}}$$

(since $(\lambda_+ - \lambda_-)^2 = (x^2 - y^2)^2 + 4(x \cdot y)^2 = |C^2|^2$). Defining the shape index

$$\boxed{\chi = \frac{|C^2|}{H}}, \quad 0 \leq \chi \leq 1,$$

$\chi = 0$ corresponds to a circular 2-frame (equal lengths, orthogonal), $0 < \chi < 1$ to an elliptic 2-frame, and $\chi = 1$ to degeneration onto a line. The motion therefore does not change the shape: it **rotates the fixed-Gram 2-frame inside $\mathbb{R}^M$** .

The square of the Gram area A_{Gram} is

$$A_{\text{Gram}}^2 = \det G = \lambda_+ \lambda_- = \frac{H^2 - |C^2|^2}{4} = \frac{H^2}{4} (1 - \chi^2),$$

with $\chi = 1 \iff \det G = 0$: the frozen class is the state whose internal 2-frame area has degenerated to zero, while $\chi = 0$ is the maximal area at fixed H .

Theorem ($\chi=1 \iff \mathbf{K}=0$). If all components are nonzero and the interaction graph (line graph) is connected, then

$$\chi = 1 \iff K = 0.$$

Proof: equality in $|C^2| = |\sum_m z_m^2| \leq \sum_m |z_m|^2 = H$ holds only when all z_m^2 share the same phase, i.e., $\theta_m \equiv \theta_0 \pmod{\pi}$, so all relative-phase sines vanish and $K = 0$. Conversely, if $K = 0$, adjacent phase differences are $0 \pmod{\pi}$ and propagate through the whole graph by connectivity. ■

Measurement (Figure 5): among the 99 runs, $\chi = 1$ (to machine precision) occurs exactly for the 5 frozen-class runs, in agreement with the analytic fact that for $m_a + m_b \equiv 0 \pmod{L}$, $z(0)$ is real times a global phase. The remaining 94 runs lie in $\chi \in [0.057, 0.486]$, with a clear gap below $\chi = 1$. By the conservation of H and C^2 , χ is a **conserved label** of each run; its time variation in the invariants series stayed within 1.3×10^{-13} over all 4097 steps.

For the two-mode initial family, χ is given in closed form in (L, m_a, m_b) . With $\alpha = \pi/L$, $M = L(L-1)/2$, and

$$S_L(r) = \sum_{\Delta=1}^{L-1} (L - \Delta) U^{r\Delta} = \begin{cases} M, & r \equiv 0 \pmod{L}, \\ \frac{LU^r}{1 - U^r} = -\frac{L}{2} + \frac{iL}{2} \cot \frac{\pi r}{L}, & r \not\equiv 0 \pmod{L}, \end{cases}$$

we have

$$C^2 = e^{-i\alpha} S_L(2m_a) + e^{i\alpha} S_L(2m_b) + 2S_L(m_a + m_b),$$

$$H = 2M + 2\Re[e^{-i\alpha} S_L(m_a - m_b)], \quad \chi(L, m_a, m_b) = \frac{|C^2|}{H}$$

(derivation in Appendix C). This closed form agrees with the measured χ , C^2 , H of all 99 runs to a maximal relative deviation of 9.3×10^{-15} , and reproduces the measured maximum $\chi(8, 3, 6) = 0.486422965736\dots$. For the frozen class $m_a + m_b \equiv 0 \pmod{L}$, the closed form gives $C^2 = H$ (real positive), so $\chi = 1$ follows analytically. Since χ is conserved, **the local Gram shape is decided by the initial integers alone, before any time evolution is executed.**

Figure 5. Catalog of the shape index χ (all 99 runs; conserved)

6.4 Local state space

The simultaneous fixing of H and C^2 defines the internal state space of one condensate:

$$\mathcal{L}_{H, C^2} = \{Z \in \mathbb{C}^M \mid Z^\dagger Z = H, Z^T Z = C^2\}.$$

For generic rank 2 ($\chi < 1$) this is the orbit of fixed-Gram 2-frames,



Figure 9: Figure 5 chi catalog

$$\mathcal{L}_{H,C^2} \simeq O(M)/O(M-2), \quad \dim_{\mathbb{R}} = 2M-3.$$

In the degenerate case ($\chi = 1$, rank 1), the dimension drops to $M-1$. Hence

$$C^2 = \text{label of the local closure}, \quad \mathcal{L}_{H,C^2} = \text{its internal local state space}.$$

The orbit $Z(0) \rightarrow Z(1) \rightarrow \dots$ is a motion inside this local space.

6.5 Embedding into zero closure

For any nonzero C^2 ,

$$(iC)^2 = -C^2,$$

so we may write

$$\sum_m z_m^2 + (iC)^2 = 0.$$

A nonzero quadratic representative therefore does not contradict the zero-closure principle. Rather, combining the many internal waves $\{z_m\}$ with the complementary component iC that represents them embeds the completed local condensate exactly into the same quadratic zero-closure form.

7. Quadratic condensation: one complex quantity represents many waves

7.1 Definition

In this paper, a subsystem in which

$$C^2 = \sum_m z_m^2$$

holds exactly is called a **condensate** in the quadratic sense.

This contraction is an identity; it uses no averaging, no low-rank SVD approximation, no normalization, and no fitting.

With respect to the quadratic quantity, the many internal degrees of freedom

$$\{z_1, z_2, \dots, z_M\}$$

are **exactly represented** by the single complex quantity

$$\boxed{C^2}.$$

C itself carries the square-root two-valuedness

$$C \leftrightarrow -C,$$

so the fundamental invariant is C^2 .

7.2 The long internal evolution belongs to the same condensate

Each run of this experiment carries the long state sequence

$$Z(0), Z(1), \dots, Z(4096).$$

The internal configuration changes substantially, but

$$Z(t)^T Z(t) = C^2$$

does not. From the higher-level quadratic readout,

$$\boxed{Z(0), Z(1), \dots, Z(4096) \text{ all belong to the same } C^2}.$$

This does not mean the 4096 states are identical — the internal states differ. But their entire history closes inside one invariant quadratic sector.

In this sense the experiment shows that

$$\boxed{\text{a long local state evolution can close inside one condensate}}.$$

8. Hierarchical closure and inter-condensate flux

8.1 Form invariance of hierarchical contraction

Consider multiple condensates. Let the internal waves of condensate n be z_{nm} and define

$$\boxed{C_n^2 = \sum_m z_{nm}^2}.$$

By the simple associativity of sums,

$$\sum_n C_n^2 = \sum_n \sum_m z_{nm}^2 = \sum_{n,m} z_{nm}^2.$$

Imposing at the higher level

$$\boxed{\sum_n C_n^2 = 0}$$

therefore automatically gives

$$\boxed{\sum_{n,m} z_{nm}^2 = 0}.$$

Grouping several C_n further,

$$D_a^2 = \sum_{n \in a} C_n^2 \implies \sum_a D_a^2 = \sum_n C_n^2.$$

Raising the hierarchy

$$z_{nm} \rightarrow C_n \rightarrow D_a \rightarrow \dots$$

never changes the form of the quadratic closure.

The central algebraic result of this paper is summarized as

$$\boxed{\text{quadratic closure is form-invariant under coarse-graining by quadratic condensation}}.$$

8.2 Antisymmetric quadratic flux between condensates

Partition the full state by condensate, $Z = (Z_1, Z_2, \dots)$, and block the real antisymmetric generator ($K_{mn} = -K_{nm}^T$). In the continuous representation $\dot{Z} = KZ$ (the flow of the per-step frozen K , the same convention as §9.2), with $q_n = Z_n^T Z_n = C_n^2$,

$$\dot{q}_n = 2Z_n^T \dot{Z}_n = 2 \sum_m Z_n^T K_{nm} Z_m,$$

and the self term vanishes by antisymmetry, $Z_n^T K_{nn} Z_n = 0$. Defining

$$\boxed{J_{nm} = 2Z_n^T K_{nm} Z_m}, \quad \boxed{J_{mn} = -J_{nm}},$$

we obtain

$$\boxed{\dot{C}_n^2 = \sum_{m \neq n} J_{nm}}, \quad \frac{d}{ds} \sum_n C_n^2 = 0.$$

Internal motion never changes its own C_n^2 ; only relations to other condensates change C_n^2 , and the exchange is antisymmetric. That is,

$$\boxed{\text{local conservation} + \text{pairwise exchange} = \text{global closure}}$$

holds as it stands.

This exchange law is not restricted to the continuous representation. Freezing $K_t = K(Z_t)$ at step t and setting $Z(s) = e^{sK_t} Z_t$ ($0 \leq s \leq \Delta\tau$), for

$$\mathcal{J}_{nm}^{(t)} = \int_0^{\Delta\tau} 2Z_n(s)^T K_{nm}^{(t)} Z_m(s) ds$$

we have exactly

$$\boxed{C_n^2(t+1) - C_n^2(t) = \sum_{m \neq n} \mathcal{J}_{nm}^{(t)}}, \quad \boxed{\mathcal{J}_{mn}^{(t)} = -\mathcal{J}_{nm}^{(t)}}.$$

The antisymmetric quadratic flux is therefore not an approximation of the continuous representation but an exact exchange law of the simultaneous update algorithm itself.

9. Minimal two-condensate system and conservative motion without assumed time

9.1 Zero closure of two condensates

As the minimal higher-level closure, consider

$$\boxed{C_1^2 + C_2^2 = 0}.$$

The nontrivial solutions are

$$C_2 = \pm i C_1.$$

The two condensates therefore have

- equal amplitude,
- a 90-degree phase difference.

This is the minimal relation of the same type as the 90-degree phase skeleton that was numerically important in the preceding study [K2]. Whether this agreement stems from the same mechanism is to be tested by the explicit construction of the two-condensate system indicated in the conclusion.

9.2 Transformations preserving the closure

For the state vector

$$\mathbf{C} = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix},$$

set

$$\mathbf{C}' = R\mathbf{C}.$$

If

$$R^T R = I,$$

then

$$\mathbf{C}'^T \mathbf{C}' = \mathbf{C}^T R^T R \mathbf{C} = \mathbf{C}^T \mathbf{C} = 0.$$

A class of state transformations that do not break the quadratic closure is thus defined.

Introducing a continuous parameter s merely as an orbit parameter,

$$R(s) = e^{sK}, \quad K^T = -K,$$

and

$$\frac{d\mathbf{C}}{ds} = K\mathbf{C},$$

with

$$\frac{d}{ds} (\mathbf{C}^T \mathbf{C}) = 0.$$

What matters is that s need not be assumed to be physical absolute time from the outset. First there is

$$\boxed{\text{closure-preserving motion between states}},$$

and s is used only to describe its orbit.

10. Local clocks: pairwise relations, not an absolute clock

Even if each condensate carries internal state evolution, no absolute clock needs to be attached to any single condensate.

Write the quadratic representative of each condensate as

$$C_n^2 = \rho_n e^{i\Theta_n}, \quad \rho_n = |C_n^2|, \quad \Theta_n \in \mathbb{R}/2\pi\mathbb{Z}.$$

$\Theta_n = \arg C_n^2$ is single-valued (the phase of C_n itself carries the two-valuedness $C \leftrightarrow -C$, §7.1).

Define the relational phase from A to B, single-valued from the quadratic representatives alone:

$$\boxed{\tilde{U}_{AB} = e^{i(\Theta_B - \Theta_A)} = \frac{C_B^2/|C_B^2|}{C_A^2/|C_A^2|}}.$$

The reverse relation is

$$\boxed{\tilde{U}_{BA} = \tilde{U}_{AB}^{-1}}.$$

Under a common phase shift of all condensates, $\Theta_n \rightarrow \Theta_n + \beta$, $\tilde{U}_{AB}$ is invariant: the absolute phase disappears completely from the relational quantity.

For three systems, the composition law

$$\boxed{\tilde{U}_{AB}\tilde{U}_{BC} = \tilde{U}_{AC}}, \quad \boxed{\tilde{U}_{AB}\tilde{U}_{BC}\tilde{U}_{CA} = 1}$$

holds immediately. If A reads B and B reads C, the clock relation from A to C composes. Local clocks are not merely pairwise ratios; they compose consistently over the complete relational network.

The directed relation of local clocks lives on the double cover of this single-valued relational quantity. Taking a lift U_{AB} with

$$\boxed{U_{AB}^2 = \tilde{U}_{AB}},$$

the reverse is

$$U_{BA} = U_{AB}^{-1}.$$

At minimal closure $C_B^2 = -C_A^2$,

$$\tilde{U}_{AB} = -1,$$

whose two lifts are $U_{AB} = +i, -i$; reversing the order of the relation exchanges $+i \leftrightarrow -i$. In the minimal two-condensate system, therefore, **the $\mathbb{Z}_2$ fiber of the double cover itself carries the direction of the local clock**. This is not a direction variable added from outside; it is intrinsic to the square-root structure that quadratic condensation possesses from the start.

No third-party absolute clock is needed here:

$$\boxed{\text{B is a clock for A, and A is a clock for B.}}$$

In this reading, time appears not as an absolute quantity attached to one system but as a directed relation between two locally closed systems.

Viewing the whole as one closed relational structure, there is only

$$\sum_n C_n^2 = 0,$$

and no single absolute clock or absolute time origin needs to be added.

Between local systems, on the other hand, the directed relations

$$A - B, \quad B - A$$

exist naturally.

What follows directly from this structure is:

$$\boxed{\text{local clocks can be defined as inter-system relations, without assuming an absolute clock.}}$$

11. Reconstruction of Kepler dynamics from relational closure

This section shows that combining the fixed Gram geometry of §6.3 and the zero-closure embedding of §6.5 with one clock-readout axiom $dt = r d\tau$ (§11.3) reconstructs the Kepler ellipse, the Kepler equation, the area law, and the inverse-square central force from a system that does not place physical time or spatial coordinates as fundamental variables.

11.1 Principal-axis representation of the fixed Gram 2-frame

For the eigenvalues $\lambda_{\pm} = (H \pm |C^2|)/2$ of §6.3, set

$$a^2 = \frac{H + |C^2|}{2}, \quad b^2 = \frac{H - |C^2|}{2}.$$

Aligning the global phase to $\arg C^2 = 0$ (a gauge choice that changes neither H nor $|C^2|$), the 2-frame takes principal-axis form. Using the complementary component of §6.5 directly, with $c_0 := \sqrt{|C^2|}$, take the three-component state

$$\mathbf{V}(0) = \begin{pmatrix} a \\ ib \\ ic_0 \end{pmatrix}.$$

Then

$$V_1^2 + V_2^2 + V_3^2 = a^2 - b^2 - c_0^2 = |C^2| - |C^2| = 0,$$

so **exact quadratic zero closure** holds. Taking the real antisymmetric generator

$$K_3 = \begin{pmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad K_3^T = -K_3,$$

the complementary component $V_3 = ic_0$ stays fixed while the first component of the closure-preserving motion $\mathbf{V}(\tau) = e^{\tau K_3} \mathbf{V}(0)$ reads out as

$$u(\tau) = V_1(\tau) = a \cos(\omega\tau) + ib \sin(\omega\tau).$$

Here τ is not external physical time but the parameter of the closure-preserving orbit. Directly from $K_3^T = -K_3$,

$$u'' = -\omega^2 u,$$

and throughout, $|V_1|^2 + |V_2|^2 = a^2 + b^2 = H$, $V_1^2 + V_2^2 = a^2 - b^2 = |C^2|$, and the total zero closure $\sum_i V_i^2 = 0$ are identically preserved.

In particular, if $C^2 = 0$ then $a = b$, the complementary component vanishes, and $e = \chi = 0$: **the minimal two-condensate zero closure of §9** ($C_2 = \pm iC_1$) **corresponds to the circular Kepler orbit**. The general elliptic orbit corresponds to $C^2 \neq 0$, i.e., to zero closure with the complementary component $i\sqrt{|C^2|}$, and any nondegenerate elliptic eccentricity $0 \leq e = \chi < 1$ is realized while exactly preserving quadratic zero closure. $\chi = 1$ is the boundary where $b = 0$, $B = 0$, and w degenerates to a segment through the focus, in agreement with the rank-1 degeneracy of §6.3.

11.2 The focal ellipse by the squaring map

Take the squaring map

$$w = u^2.$$

Setting $E = 2\omega\tau$,

$$w = D + A \cos E + iB \sin E, \quad A = \frac{H}{2}, \quad D = \frac{|C^2|}{2}, \quad B = \frac{1}{2} \sqrt{H^2 - |C^2|^2},$$

and

$$A^2 - B^2 = D^2$$

holds ($A^2 - B^2 = |C^2|^2/4 = D^2$). The ellipse therefore has center D , semi-major axis A , and focal distance $c_f = \sqrt{A^2 - B^2} = D$, so **one focus coincides with the origin**. The eccentricity is

$$e = \frac{D}{A} = \frac{|C^2|}{H} = \chi.$$

The shape index χ is precisely the eccentricity of the Kepler ellipse after the squaring map.

The distance from the focus (origin) is

$$\boxed{r = |w| = |u|^2}, \quad r = A(1 + e \cos E).$$

To match the usual Kepler convention (pericenter origin), set $E_K = E - \pi$:

$$r = A(1 - e \cos E_K).$$

11.3 Clock-readout axiom

In this paper the readout of the relational clock is posited as one **axiom**:

$$\boxed{dt = r d\tau = |u|^2 d\tau}$$

(of the same form as a Sundman-type transformation). This paper does not derive this readout rule.

Note that the hyperbolic local closure $r^2 - c^2 t^2 = R^2$ yields $c dt = r d\eta$ via $r = R \cosh \eta$, $ct = R \sinh \eta$, but the hyperbolic angle η cannot be identified with the orbit parameter τ : on a Kepler orbit $r(\tau) = A(1 + e \cos 2\omega\tau)$ is bounded and periodic, whereas $\eta = \gamma\tau$ would give the unbounded $r = R \cosh(\gamma\tau)$, incompatible with §11.2 (recorded here as part of the investigation). The hyperbolic closure is therefore kept as a separate relational closure, and η and τ are not identified. Deriving the readout rule $dt/d\tau = |u|^2$ from the relational clock $\tilde{U}_{AB}$ or from the quadratic closure structure is an open problem (§15.2).

11.4 The Kepler equation

From $E = 2\omega\tau$, $d\tau = dE/(2\omega)$, so

$$dt = \frac{A}{2\omega}(1 + e \cos E) dE,$$

and integrating,

$$t - t_0 = \frac{A}{2\omega}(E + e \sin E).$$

Moving to the pericenter convention $E_K = E - \pi$ with pericenter passage time $t_p = t_0 + \pi A/(2\omega)$,

$$\boxed{M = E_K - e \sin E_K}, \quad M = n(t - t_p), \quad \boxed{n = \frac{2\omega}{A}}.$$

In a system that did not place external physical time as a fundamental variable, the Kepler equation appears under the clock-readout axiom $dt = r d\tau$.

11.5 The area law (Kepler's second law)

From $w = u^2$ and $dt = |u|^2 d\tau$,

$$\dot{w} = \frac{dw/d\tau}{dt/d\tau} = \frac{2uu'}{|u|^2} = \frac{2u'}{\bar{u}}.$$

The specific angular momentum about the focus at the origin, $h = \Im(\bar{w}\dot{w})$, satisfies

$$h = 2\Im(\bar{u}u'), \quad \Im(\bar{u}u') = ab\omega,$$

so

$$\boxed{h = 2ab\omega = \text{const}}, \quad \boxed{\frac{dA_{\text{swept}}}{dt} = \frac{h}{2} = ab\omega = \text{const}}.$$

This is Kepler's second law.

11.6 The inverse-square central force and Kepler's third law

Use the oscillator-side conserved quantity $E_{\text{osc}} = \frac{1}{2}(|u'|^2 + \omega^2|u|^2)$. Differentiating $\dot{w} = 2u'/\bar{u}$ once more with respect to t , and using $u'' = -\omega^2 u$ and $r = |w| = |u|^2$,

$$\ddot{w} = \frac{2}{|u|^2} \left[\frac{u''}{\bar{u}} - \frac{u'\bar{u}'}{\bar{u}^2} \right] = -\frac{2}{u\bar{u}^3} (\omega^2 u\bar{u} + u'\bar{u}') = -\frac{4E_{\text{osc}}}{u\bar{u}^3},$$

that is,

$$\boxed{\ddot{w} = -\frac{4E_{\text{osc}}}{|w|^3} w}, \quad \boxed{\ddot{\mathbf{r}} = -\mu \frac{\mathbf{r}}{r^3}}, \quad \boxed{\mu = 4E_{\text{osc}}}.$$

On the elliptic solution, $E_{\text{osc}} = \frac{1}{2}\omega^2(a^2 + b^2) = \frac{1}{2}\omega^2 H$, so

$$\boxed{\mu = 2\omega^2 H}.$$

Moreover, from $A = H/2$ and $n = 2\omega/A$,

$$\boxed{n^2 A^3 = 4\omega^2 A = 2\omega^2 H = \mu},$$

consistent with Kepler's third law.

All formulas of this section were verified independently both symbolically and numerically (the focal ellipse form, $e = \chi$, the Kepler equation, the constancy of h , the residual of $\ddot{w} + \mu w/r^3$, and $n^2 A^3 = \mu$, all within numerical-differentiation accuracy).

12. Inflation-type transitions: the two extremes sampled by this survey

12.1 The indicator of departure from the initial plane

With the same definition as the preceding inflation experiments, let E_0 be the real two-dimensional plane spanned by the real and imaginary parts of the initial state Z_0 , and let $Z_{\perp}(t)$ be the orthogonal component:

$$f(t) = \frac{\langle Z_{\perp}, Z_{\perp} \rangle}{\langle Z, Z \rangle}.$$

Figure 6a. Inflation indicator $H_{\perp}/H$: main 72

Figure 6b. Inflation indicator $H_{\perp}/H$: gcd + den=40 controls

In all 99 runs, $f(0)$ lies essentially at the display floor by construction. Afterwards,

- 5 runs remained at their fixed points,
- 94 runs reached $O(1)$ within the first few steps.

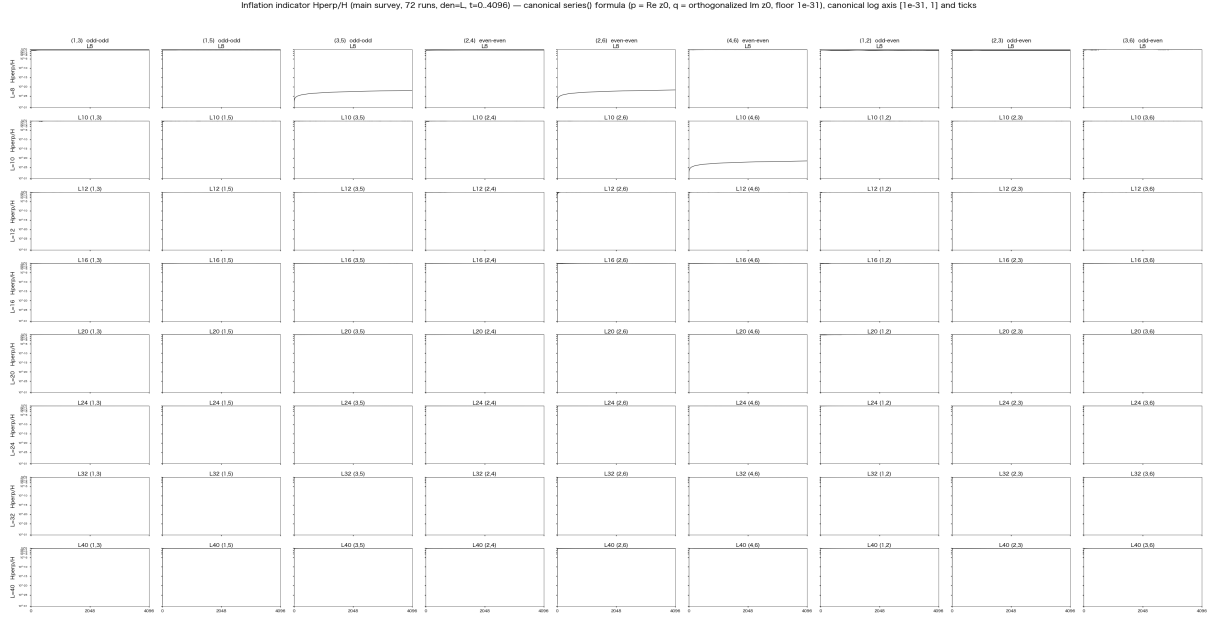


Figure 10: Figure 6a H_{perp}/H , main 72

That is, the relaxation curve seen in the preceding study [K2],

$$\text{floor} \rightarrow \text{onset} \rightarrow \text{saturation},$$

was not contained in a single one of the present 99 runs.

Note that, from the fixed Gram geometry of §6.3, $H_{\perp}/H$ measures not a growth of energy but **the degree to which the fixed-Gram 2-frame rotates and develops out of the initial 2-plane**.

12.2 Present hypotheses

What this survey establishes is that the 99 runs contain only the two sides of the relaxation curve (exact fixed points and immediate saturation). The generation condition is to be narrowed by comparing the following three regions.

1. **The over-symmetric extreme:** exact fixed points satisfying the conjugate-pair condition.
2. **The low-symmetry extreme:** the 94 runs departing immediately from the initial plane.
3. **In between:** the region with a finite floor stay and onset observed in past high-symmetry 90-degree systems.

Also, the present fixed points have comparatively small centroids, but small centroids alone do not freeze a run. What decides freezing is $\chi = 1$ (the theorem of §6.3); in the measurements too, only the 5 fixed runs have $\chi = 1$ while the other 94 have $\chi \leq 0.486$ (Figure 5). Hence the candidate combination is

$$\boxed{\text{high symmetry} + \text{centroid near the origin} + \text{yet not an exact fixed point}},$$

to be verified by the perturbation-series experiments indicated in the conclusion.

13. On the “arrow of time” and entropy

In the basic structure of this paper, entropy need not be introduced to define the direction of time.

Given a two-body system A, B , the directed relation

Inflation indicator H_{perp}/H (gcd series 18 runs, top 6 rows + den=40 controls 9 runs, bottom 3 rows)
 Canonical series() formula, floor $1e-31$, canonical log axis [$1e-31$, 1] and ticks

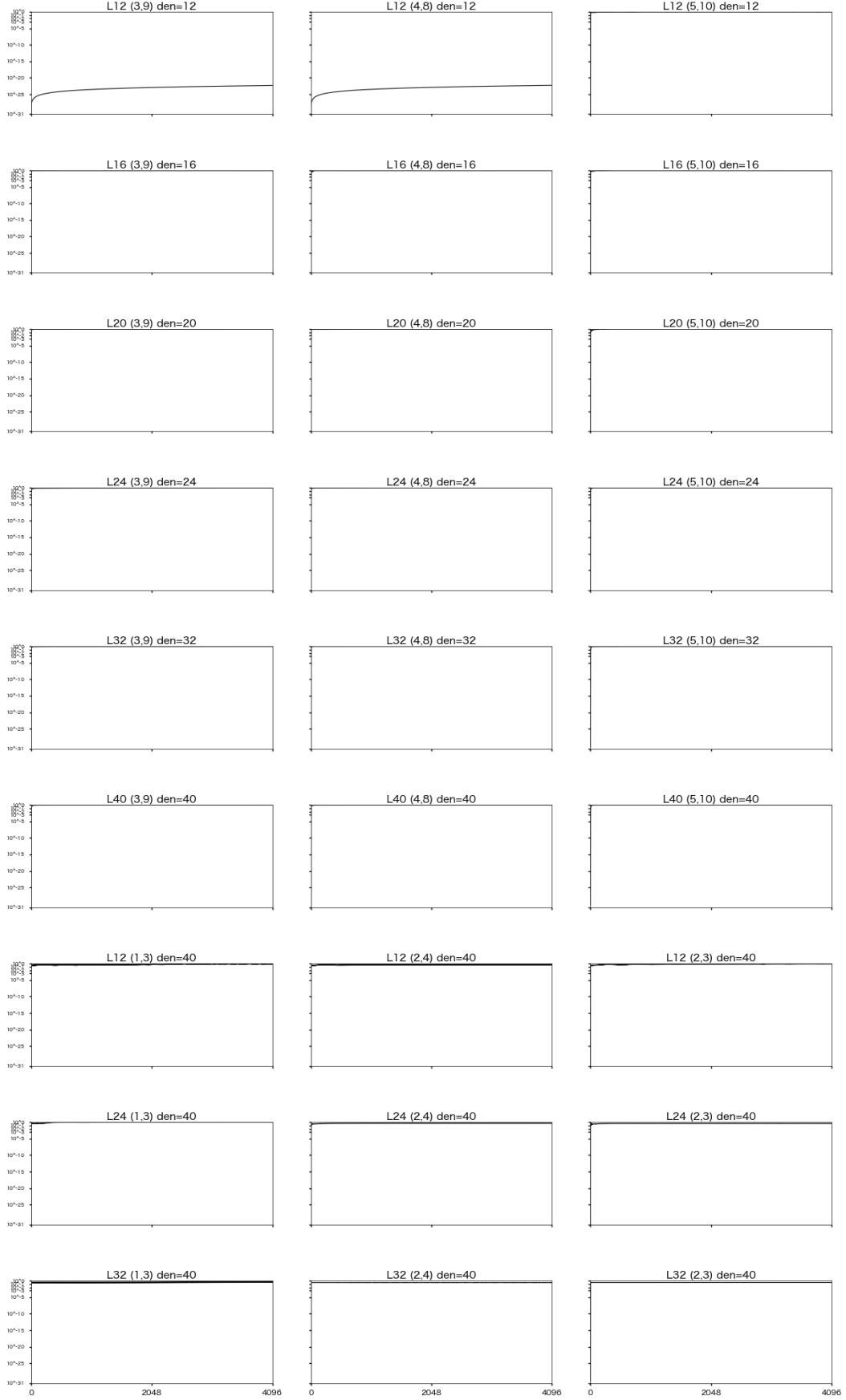


Figure 11: Figure 6b H_{perp}/H , gcd+control

$$\tilde{U}_{BA} = \tilde{U}_{AB}^{-1}$$

exists from the start. The direction of the local clock can therefore be defined by the pairwise relation. Describing the whole from outside, on the other hand, there is only the closed structure

$$\sum_n C_n^2 = 0,$$

and no absolute arrow of time needs to be placed as a fundamental variable.

If entropy growth is observed under macroscopic coarse-graining, it may correspond to state reorganization along the local relational clocks. This paper does not assume

$$\boxed{\text{entropy growth creates the direction of time}};$$

rather it leaves as a future topic

$$\boxed{\text{local relational evolution comes first; the entropy direction may be its coarse-grained image}}.$$

This is not a mathematical main result of this paper but a candidate physical interpretation arising from the structure of quadratic condensation and local clocks.

14. Relation to prior work

The quadratic condensation and hierarchical closure of this study were not derived by assuming the following external works. We first found

$$\sum_m z_m^2 = C^2$$

from the numerical experiments and the algebraic structure, and only afterwards examined the conceptual relation to existing studies that do not take external time as a fundamental variable.

Page and Wootters [E1] treat the closed total system as a stationary state and describe observed evolution through correlations with an internal clock. The common point with our work is that no external absolute clock is fundamental. In contrast, we do not start from a heterogeneous clock/system split: homogeneous local condensates A and B read each other as clocks.

Rovelli [E2] discussed describing observable evolution even where a fundamental time variable is not defined. Our study does not start from the canonical formalism of quantum gravity but from the quadratic closure of finite complex relational waves and real orthogonal conservative motion.

Boette and Rossignoli [E3] treat finite-dimensional quantum clocks and discrete history states, including multiple instants in one static structure. The contact point is that a long state sequence can be treated as the interior of one closed structure; however, we assume no independent clock Hilbert space.

Wharton [E4] argued against taking the sequential generation of the future from initial conditions as the only fundamental form, using time-symmetric boundary conditions constraining whole histories. The contact point is that no absolute future-generating direction needs to be placed on the side of the total system; the difference is that our starting point is finite quadratic closure and relational phase motion, not an action principle or continuous-field boundary conditions.

The correspondence between the harmonic oscillator and the Kepler problem via the squaring map is itself the known correspondence going back to Bohlin [E5] and Levi-Civita [E6]. This study did not construct the model by assuming that known transformation: the squaring map appeared independently from relational closure and fixed Gram geometry, and the comparison with the known correspondence was

made after the results were obtained. The difference is that in our study the time variable t itself is not given externally but appears as the relational clock via the clock-readout axiom $dt = r d\tau$.

15. What this paper establishes / does not establish

15.1 Established mathematical and numerical results

1. In the present dynamics $K^T = -K$, and each step's update $R = e^{\Delta\tau K}$ is real orthogonal.
2. Therefore

$$Z^T Z = \sum_m z_m^2 = C^2$$

is exactly conserved.

3. The 99-run, 4096-step data confirmed this conservation within numerical precision.
4. The many waves contract exactly to one C^2 with respect to the quadratic quantity.
5. For multiple condensates, with

$$C_n^2 = \sum_m z_{nm}^2,$$

the identity

$$\sum_n C_n^2 = \sum_{n,m} z_{nm}^2$$

holds exactly.

6. The nontrivial solutions of the two-condensate zero closure

$$C_1^2 + C_2^2 = 0$$

have a 90-degree phase difference.

7. The closure-preserving motion is defined by real orthogonal transformations, and in the continuous representation by real antisymmetric generators.
8. The simultaneous conservation of H and C^2 is equivalent to fixing the Gram matrix of the real and imaginary parts, with eigenvalues

$$\lambda_{\pm} = \frac{H \pm |C^2|}{2}.$$

The shape index $\chi = |C^2|/H \in [0, 1]$ is a conserved label of each run.

9. Under nonvanishing components and line-graph connectivity, $\chi = 1 \iff K = 0$. In the measurements, $\chi = 1$ occurs exactly for the 5 frozen runs (all others have $\chi \leq 0.486$, Figure 5).
10. For a condensate partition, $J_{nm} = 2Z_n^T K_{nm} Z_m$ satisfies $J_{mn} = -J_{nm}$ and

$$\dot{C}_n^2 = \sum_{m \neq n} J_{nm}, \quad \frac{d}{ds} \sum_n C_n^2 = 0$$

(the self term vanishes).

11. The local state space is, generically ($\chi < 1$), $\mathcal{L}_{H,C^2} \simeq O(M)/O(M-2)$ of real dimension $2M-3$ (and $M-1$ for $\chi = 1$).
12. The relational phase $\tilde{U}_{AB} = e^{i(\Theta_B - \Theta_A)}$ ($\Theta_n = \arg C_n^2$) is single-valued and common-phase invariant, and satisfies

$$\tilde{U}_{BA} = \tilde{U}_{AB}^{-1}, \quad \tilde{U}_{AB}\tilde{U}_{BC} = \tilde{U}_{AC}.$$

The double-cover lift $U_{AB}^2 = \tilde{U}_{AB}$ gives $U_{AB} = \pm i$ at minimal closure, with the $\mathbb{Z}_2$ fiber carrying the direction.

13. The χ of the two-mode initial family is given by the closed form in (L, m_a, m_b) (via $S_L(r)$, Appendix C) and agrees with the measurements of all 99 runs to machine precision.
14. Under the squaring map $w = u^2$, the fixed Gram ellipse maps to an ellipse with a focus at the origin whose eccentricity is $e = \chi = |C^2|/H$.
15. Under the clock-readout axiom $dt = r d\tau = |u|^2 d\tau$, the Kepler equation $M = E_K - e \sin E_K$ (with $n = 2\omega/A$) is derived (the axiom itself is not derived).
16. The same transformation yields the constancy of the areal velocity $h = 2ab\omega$ and the inverse-square central force $\ddot{\mathbf{r}} = -\mu\mathbf{r}/r^3$ (with $\mu = 2\omega^2 H$ and $n^2 A^3 = \mu$).

15.2 Not yet established

1. Which actual particles, fields, or spatial regions the C_n correspond to.
 2. That the 90-degree phase relation is identical to actual spacetime structure.
 3. That the local clock obtained from relational phase agrees with physical proper time.
 4. The necessary and sufficient conditions for inflation-type relaxation curves.
 5. That high symmetry and near-origin centroid conditions guarantee relaxation-type evolution.
 6. That the entropy direction necessarily follows the direction of local relational clocks.
 7. That this model fully reproduces the time concept of general relativity or standard quantum theory.
 8. The derivation of the clock-readout axiom $dt = r d\tau$ from the relational clock $\tilde{U}$ or from the quadratic closure structure.
-

16. Conclusion

This study extended the minimal two-wave closed system of paper 1 [K1] to many-wave complete relational dynamics. The anticipated relaxation-type temporal structure was not obtained in the 99 runs. From this outcome, a simpler and stronger invariant structure became clear:

$$\boxed{\sum_m z_m^2 = C^2}.$$

Since it can be written as

$$\sum_m z_m^2 + (iC)^2 = 0,$$

the many internal waves are exactly gathered, with respect to the quadratic quantity, into a single complex condensed quantity C^2 .

For multiple condensates,

$$C_n^2 = \sum_m z_{nm}^2$$

can be defined, and

$$\boxed{\sum_n C_n^2 = 0}$$

can again be used, in the same form, as the higher-level closure. Quadratic closure therefore loses no form under condensing the internal degrees of freedom and raising the hierarchy.

The minimal two-condensate system

$$C_1^2 + C_2^2 = 0$$

carries the 90-degree phase relation, and the closure-preserving motion can be defined without assuming external absolute time. Given two local condensates A and B, the phase difference from A to B and that from B to A exist as sign-reversed relations of each other. Hence, without placing an absolute clock, the relational local clock

$$\boxed{\text{B is a clock for A, and A is a clock for B}}$$

can be constructed.

Viewing the whole at once, neither an absolute clock nor an absolute arrow of time needs to be introduced as a fundamental variable; between local systems there are directed relations from which local order can be read.

The most compact conclusion of this paper is the chain

$$K^T = -K \implies (H, C^2) \text{ conserved} \implies \text{fixed-Gram local space} \implies \text{quadratic condensation and hierarchical closure} \implies \dots$$

The central result of this paper is that, from finite relational closure that does not assume physical time and spatial coordinates as fundamental variables, conserved geometry, local clocks, the Kepler equation, and the inverse-square central force are connected through one quadratic structure.

The next-stage task is to construct multiple condensates preserving quadratic closure explicitly, and to test independently whether the past inflation-type relaxation curves are reproduced under high-symmetry 90-degree configurations with near-origin centroids.

References

Self-citations

[K1] N. Kihara, “A Finite Complete-Relation Closed System with Time Evolution as Vertices — Exact Minimal Closure at $N = 1, K = 4$ under $U^K = I$ and Complete-Network Analysis of Even and Odd Harmonics,” 2026. Version DOI: 10.5281/zenodo.22763330; Concept DOI: 10.5281/zenodo.22763329.

[K2] N. Kihara, “Mechanism of Inflation-like Rapid Expansion in a Self-Consistent Closed System of Relational Waves,” 2026. Version DOI: 10.5281/zenodo.22176949; Concept DOI: 10.5281/zenodo.22112008.

External references: conceptual neighbors

[E1] D. N. Page and W. K. Wootters, “Evolution without evolution: Dynamics described by stationary observables,” *Physical Review D* **27**, 2885–2892 (1983). DOI: 10.1103/PhysRevD.27.2885.

[E2] C. Rovelli, “Time in quantum gravity: An hypothesis,” *Physical Review D* **43**, 442–456 (1991). DOI: 10.1103/PhysRevD.43.442.

[E3] A. Boette and R. Rossignoli, “History states of systems and operators,” *Physical Review A* **98**, 032108 (2018). DOI: 10.1103/PhysRevA.98.032108.

[E4] K. Wharton, “Time-Symmetric Boundary Conditions and Quantum Foundations,” *Symmetry* **2**, 272–283 (2010). DOI: 10.3390/sym2010272.

[E5] K. Bohlin, “Note sur le problème des deux corps et sur une intégration nouvelle dans le problème des trois corps,” *Bulletin Astronomique* **28**, 113–119 (1911).

[E6] T. Levi-Civita, “Sur la régularisation du problème des trois corps,” *Acta Mathematica* **42**, 99–144 (1920).

Appendix A. List of all figure files used

This paper cites **only the completed figures** from the figure catalogs created on 2026-09-16 (the intermediate versions v1–v3 of the initial-state catalog are not cited). The English-labeled figures below were generated by faithful copies of the original programs, changing only label strings and output filenames.

1. fig_initial_states_all99_v4_page1_main72_en_20260916.png
2. fig_initial_states_all99_v4_page2_gcd18_ctrl9_en_20260916.png
3. fig_waveforms_linear_sum_all99_page1_main72_en_20260916.png
4. fig_waveforms_linear_sum_all99_page2_gcd18_ctrl9_en_20260916.png

5. `fig_final_states_all99_page1_main72_en_20260916.png`
6. `fig_final_states_all99_page2_gcd18_ctrl9_en_20260916.png`
7. `fig_residual_timeseries_all99_page1_main72_en_20260916.png`
8. `fig_residual_timeseries_all99_page2_gcd18_ctrl9_en_20260916.png`
9. `fig_inflation_hperp_all99_page1_main72_en_20260916.png`
10. `fig_inflation_hperp_all99_page2_gcd18_ctrl9_en_20260916.png`
11. `fig_chi_all99_en_20260916.png`

Figures 1–4 (a/b) correspond to items 1–8, Figures 6a/b to items 9–10, and Figure 5 (the χ catalog) to item 11.

Appendix B. Reproducibility notes

The 99-run raw-trajectory database of this paper stores 4097 states per run together with QA, metadata, invariants, and SHA256 records. All 99 runs were fixed after the collection was completed, and the figures of this paper were generated by read-only access to the stored `states.npz`. No trajectory was rerun and no condition was changed at figure-generation time.

The central conservation law used in this paper,

$$\sum_m z_m^2 = C^2,$$

is not a hypothesis introduced by reinterpretation after the figures were made; it is the exact algebraic conservation law inherent in the real antisymmetric generator and the real orthogonal update. What is new is the rereading of this conserved quantity not as a “residual from zero” but as the “condensed quantity representing the many waves”.

Appendix C. Closed-form derivation of $\chi(\mathbf{L}, \mathbf{m_a}, \mathbf{m_b})$

For the initial condition $z_{jk}(0) = c_a U^{m_a \Delta} + c_b U^{m_b \Delta}$ ($c_a = e^{-i\pi/(2L)}$, $c_b = e^{+i\pi/(2L)}$, $\Delta = k - j$), the number of edges with difference $\Delta = 1, \dots, L - 1$ is $L - \Delta$. Define

$$S_L(r) := \sum_{\Delta=1}^{L-1} (L - \Delta) U^{r\Delta}, \quad U = e^{2\pi i/L}.$$

Evaluation of $S_L(r)$. If $r \equiv 0 \pmod{L}$, then $U^{r\Delta} = 1$ and $S_L(r) = \sum_{\Delta=1}^{L-1} (L - \Delta) = M$. If $r \neq 0$, then for $\omega = U^r \neq 1$ with $\omega^L = 1$,

$$\sum_{\Delta=1}^{L-1} \omega^\Delta = -1, \quad \sum_{\Delta=1}^{L-1} \Delta \omega^\Delta = \frac{L}{\omega - 1}$$

(the latter is the standard sum using $\omega^L = 1$). Therefore

$$S_L(r) = L \cdot (-1) - \frac{L}{\omega - 1} = -L \frac{\omega}{\omega - 1} = \frac{L\omega}{1 - \omega}.$$

With $\omega = e^{i\theta}$ ($\theta = 2\pi r/L$), $\frac{\omega}{1 - \omega} = -\frac{1}{2} + \frac{i}{2} \cot \frac{\theta}{2}$, so

$$S_L(r) = -\frac{L}{2} + \frac{iL}{2} \cot \frac{\pi r}{L}.$$

C^2 and H . Using $c_a^2 = e^{-i\alpha}$, $c_b^2 = e^{i\alpha}$, $c_a c_b = 1$, $c_a \bar{c}_b = e^{-i\alpha}$ ($\alpha = \pi/L$), and collecting edge sums by Δ class,

$$C^2 = \sum_e z_e^2 = e^{-i\alpha} S_L(2m_a) + e^{i\alpha} S_L(2m_b) + 2S_L(m_a + m_b),$$

$$H = \sum_e |z_e|^2 = 2M + e^{-i\alpha} S_L(m_a - m_b) + e^{i\alpha} S_L(m_b - m_a) = 2M + 2\Re[e^{-i\alpha} S_L(m_a - m_b)]$$

(using $S_L(-r) = \overline{S_L(r)}$). Hence

$$\chi(L, m_a, m_b) = \frac{|C^2|}{H}.$$

Frozen class. If $m_a + m_b \equiv 0 \pmod{L}$, then $S_L(m_a + m_b) = M$ and $m_a - m_b \equiv 2m_a \pmod{L}$, so

$$C^2 = 2M + 2\Re[e^{-i\alpha} S_L(2m_a)] = H,$$

i.e., C^2 is real positive and $\chi = 1$ follows analytically.

Numerical check. This closed form agrees with the measured χ , C^2 , H of all 99 runs to a maximal relative deviation of 9.3×10^{-15} .